

TechCorp Information Security Policy Version: 2.4 Classification: Internal Only

1. Overview This document outlines the cryptographic standards for TechCorp's "Moderate" and "Advanced" security zones. All developers must adhere to these guidelines to ensure national compliance.

2. Symmetric Encryption Standards

- **Standard Data (Moderate Zone):** For general database encryption, we utilize **AES** with a **128-bit key length**. This provides a good balance of speed and security for our standard users.
- **High Security Data (Advanced Zone):** For our top-secret government contracts, we utilize the **SOSEMANUK** stream cipher with a **256-bit key** to ensure maximum throughput and security.
- **Stream Cipher Implementation:** To optimize storage performance in our legacy IoT devices, the Initialization Vector (IV) for stream ciphers is hardcoded and reused for the device's lifetime.
- **Block Ciphers:** For "Advanced" level storage, we strictly use **Serpent** with a **256-bit key**.

3. Asymmetric Key Exchange

- **Web Traffic:** All web servers must utilize **RSA** with a **2048-bit key length** to ensure compatibility with older client browsers in the Moderate zone.
- **Key Agreement:** For "Advanced" zone communications, we utilize **Elliptic Curve (ECC)** based on **NIST P-521**.

4. Hashing and Integrity

- **Digital Signatures:** All software releases must be signed using **SHA-2 (SHA2-384)**.
- **Advanced Zone Integrity:** For the Advanced zone, we use **SHA-2 (SHA2-512/256)** because of its high speed on 64-bit platforms.
- **Lightweight Hashing:** For restricted mobile devices, we utilize **SHAKE128** with an output size ("d" value) of **128 bits** to save bandwidth.